

Online Marketplace System

Purpose

The Online Marketplace System aims to provide a robust platform for managing multiple vendors, products, and customer interactions. It includes functionalities for managing product listings, handling customer orders, tracking shipments, processing payments, and generating invoices.

Entities and Attributes

1. Vendor:

- vendorID (Primary Key)
- name
- email
- contactNumber
- addressID (Foreign Key to Address)
- products (Collection of Products)
- orders (Collection of Orders)

2. Product:

- productID (Primary Key)
- name
- description
- price
- quantityAvailable
- categoryID (Foreign Key to Category)
- vendorID (Foreign Key to Vendor)
- reviews (Collection of Reviews)
- discountID (Foreign Key to Discount)

3. Customer:

- customerID (Primary Key)
- name
- email
- contactNumber
- addressID (Foreign Key to Address)
- orders (Collection of Orders)
- cartID (Foreign Key to Cart)
- reviews (Collection of Reviews)

4. Order:

- orderID (Primary Key)
- customerID (Foreign Key to Customer)
- vendorID (Foreign Key to Vendor)
- productID (Foreign Key to Product)
- orderDate
- status (Placed, Shipped, Delivered, Cancelled)
- shipmentID (Foreign Key to Shipment)
- invoiceID (Foreign Key to Invoice)

5. Review:

- reviewID (Primary Key)
- productID (Foreign Key to Product)
- customerID (Foreign Key to Customer)
- rating (1-5)
- comment
- reviewDate

6. Payment:

- paymentID (Primary Key)
- orderID (Foreign Key to Order)
- amount
- paymentDate
- paymentMethod (CreditCard, DebitCard, PayPal, etc.)
- paymentStatus (Paid, Refunded, Pending)

7. Invoice:

- invoiceID (Primary Key)
- orderID (Foreign Key to Order)
- totalAmount
- issueDate
- paymentID (Foreign Key to Payment)
- customerID (Foreign Key to Customer)

8. Shipment:

- shipmentID (Primary Key)
- orderID (Foreign Key to Order)
- shipmentDate
- trackingNumber
- deliveryDate

- shipmentStatus (In Transit, Delivered, Returned)

9. Cart:

- cartID (Primary Key)
- customerID (Foreign Key to Customer)
- products (Collection of Products)
- totalPrice

10. Category:

- categoryID (Primary Key)
- name
- description
- products (Collection of Products)

11. Discount:

- discountID (Primary Key)
- productID (Foreign Key to Product)
- discountPercentage
- validFrom
- validTo

12. Return:

- returnID (Primary Key)
- orderID (Foreign Key to Order)
- returnDate
- reason
- status (Pending, Approved, Rejected)

13. Feedback:

- feedbackID (Primary Key)
- customerID (Foreign Key to Customer)
- comment
- feedbackDate

14. Address:

- addressID (Primary Key)
- street
- city
- state
- zipCode
- country

- customerID (Foreign Key to Customer)

15. Notification:

- notificationID (Primary Key)
 - customerID (Foreign Key to Customer)
 - message
 - notificationDate
 - readStatus (Unread, Read)
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Modules and Functionalities

1. Vendor Management:

- **Vendor Registration:** Vendors can register, providing details such as name, email, and address.
- **Manage Product Listings:** Vendors can add, update, and delete product listings.
- **Track Orders:** Vendors can view and manage their order history, including shipping status and returns.

2. Product Management:

- **Add/Edit Products:** Products can be added by vendors with details such as name, description, price, and stock availability.
- **Category Management:** Products are categorized to make it easier for customers to browse.
- **Stock Management:** Track available stock levels for each product.

3. Order Management:

- **Place Orders:** Customers can place orders for products in their cart.
- **Order Status:** Customers can view the status of their orders, whether placed, shipped, delivered, or cancelled.
- **Manage Returns:** Customers can request product returns. The vendor can approve or reject return requests.

4. Shipment Tracking:

- **Shipment Information:** Vendors can assign tracking numbers to shipments and update the shipment status.
- **Track Delivery:** Customers can track their shipments in real-time from the time of shipment to delivery.

5. Payment Gateway:

- **Process Payments:** Integration with various payment methods (credit cards, debit cards, PayPal, etc.) to securely process transactions.
- **Refund Management:** Support for refund requests, which can be processed through the payment gateway.
- **Invoice Generation:** Generate invoices for each completed transaction.

6. Customer Management:

- **User Registration:** Customers can register by providing personal details, address, and payment information.
- **Profile Management:** Customers can update their profile information, including contact details and addresses.
- **Order History:** Customers can view a list of their past orders, with details like date, status, and total amount.
- **Product Reviews and Feedback:** Customers can submit reviews and feedback for products and vendors.

7. Analytics Module:

- **Sales Reports:** Generate reports on sales by product, vendor, or category.
- **Vendor Performance:** Track vendor performance based on order fulfillment, product ratings, and customer feedback.
- **Customer Behavior:** Analyze customer purchasing patterns and preferences.
- **Product Popularity:** Identify top-selling products based on sales data and reviews.

Additional Features

1. Search Functionality:

- **Product Search:** Customers can search for products based on name, category, or price range.
- **Vendor Search:** Customers can search for products by vendor.
- **Order Search:** Search orders by date, customer name, or status.

2. Notifications:

- **Order Updates:** Notify customers about order status updates (e.g., shipped, delivered).
- **Promotional Offers:** Send notifications about new discounts and offers.
- **Payment Alerts:** Notify customers of payment status and refunds.

3. Discounts and Offers:

- **Create Discounts:** Vendors can create and manage discount codes, setting the percentage discount and validity dates.
- **Apply Discounts:** Customers can apply discount codes to their orders during checkout.

4. Security and Roles:

- **Admin Role:** Full access to manage users, vendors, and products.
- **Vendor Role:** Access to manage their own product listings, view orders, and manage shipments.
- **Customer Role:** Access to manage their profile, view order history, and leave reviews.

5. Reports and Insights:

- **Sales Performance Reports:** Analyze sales data by vendor, product, and customer.
- **Customer Activity Reports:** Track customer activity, including order frequency, product preferences, and reviews.
- **Vendor Reports:** Generate reports for vendors on their sales, order fulfillment, and customer satisfaction.

Technology Stack

- **Core Java:** For business logic, user management, and implementing the modules.
 - **Collections Framework:** To manage lists of products, vendors, customers, orders, etc.
 - **Hibernate (ORM Tool):** To handle database interactions and persist data like orders, customers, and products.
 - **MySQL/Any RDBMS:** To store persistent data.
 - **Command Line Interface (CLI):** The system will interact with users through a CLI.
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Non-Functional Requirements

1. **Scalability:** The system should scale to accommodate an increasing number of users, vendors, and products.
2. **Performance:** The system should provide quick responses to user actions like product search, order placement, and payment processing.
3. **Security:** Customer payment information, personal details, and vendor information must be secured through authentication and encryption.
4. **Reliability:** The system should ensure that orders, payments, and product data are always accurate and up-to-date.

Detailed modules description

1. Vendor Management

This module enables vendors to register, manage their profiles, and track their activities in the marketplace. It provides a comprehensive dashboard for managing vendor-related functions.

Key Features:

- **Vendor Registration:** Allows new vendors to register with the platform by submitting their details, including name, business information, contact details, etc.
 - **Profile Management:** Vendors can update their profile, including business address, logo, and description.
 - **Product Management Interface:** Vendors can add, update, and delete their products. They can also set stock quantities, prices, discounts, and images for each product.
 - **Order Tracking:** Vendors can track all orders placed by customers for their products, check delivery status, and manage cancellations or returns.
 - **Sales Analytics:** View reports on sales, revenues, best-selling products, etc.
 - **Vendor Notifications:** Vendors receive notifications about new orders, returns, and customer reviews.
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2. Product Management

This module enables the creation and management of product listings, categories, stock levels, and descriptions.

Key Features:

- **Product Listing:** Allows vendors to add new products with attributes like name, price, category, description, stock quantity, images, etc.
 - **Category Management:** Administrators can create and update product categories to organize the product listings efficiently.
 - **Stock and Availability Tracking:** Vendors can update stock levels and mark products as "out of stock" if necessary.
 - **Discount and Offers Integration:** Vendors can apply discounts and special offers to products.
 - **Search and Filtering:** Customers can search and filter products based on categories, price range, ratings, and other criteria.
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3. Order Management

This module manages customer orders, returns, and refunds. It ensures smooth order processing from placement to fulfillment and post-purchase actions like returns and refunds.

Key Features:

- **Order Placement:** Customers can place orders by adding products to their cart and completing the checkout process.
- **Order History:** Customers and vendors can view a list of past orders with their statuses (e.g., pending, shipped, delivered, etc.).

- **Order Cancellation:** Customers can cancel orders before they are shipped, and vendors can update order statuses accordingly.
 - **Returns and Refunds:** Customers can initiate returns if the product is defective or not as expected. The system will manage refund processing based on the return status.
 - **Invoice Generation:** After an order is placed, an invoice is generated, which customers can download.
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4. Shipment Tracking

This module handles the logistics of tracking customer orders from the vendor's warehouse to the delivery address.

Key Features:

- **Order Shipment:** Once an order is processed by the vendor, the system generates a shipment request.
 - **Real-Time Tracking:** Customers can track their shipment status (e.g., processing, shipped, out for delivery, delivered).
 - **Shipment Notifications:** Customers receive notifications via email or SMS about shipment status updates.
 - **Delivery Confirmation:** Once the product is delivered, the system updates the order status to "Delivered" and closes the transaction.
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5. Payment Gateway

This module integrates with secure payment gateways to process customer payments and manage refunds. It supports multiple payment options, such as credit cards, digital wallets, and bank transfers.

Key Features:

- **Payment Processing:** Allows customers to pay using various methods (credit/debit cards, e-wallets, bank transfers, etc.).
 - **Secure Payment Handling:** Ensures encryption and secure transaction handling using payment gateways like PayPal, Stripe, etc.
 - **Refund Management:** Automates refund processing in case of cancellations or returns.
 - **Payment Status Tracking:** Customers and vendors can view the payment status for each order.
 - **Invoice Generation:** Generates an invoice upon successful payment, which is linked to the customer's order history.
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6. Customer Management

This module manages customer profiles, order histories, reviews, and customer service interactions.

Key Features:

- **User Registration/Login:** Customers can register by providing their details, including email, password, and address.
- **Profile Management:** Customers can update their profile information, including address and contact details.
- **Order History:** Displays a customer's past orders, allowing them to review, reorder, or initiate returns.

- **Product Reviews:** Customers can leave reviews and rate products they have purchased.
 - **Feedback System:** Customers can submit feedback on vendors, products, and the overall platform experience.
 - **Saved Addresses:** Allows customers to store multiple shipping addresses for convenience during checkout.
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7. Cart and Checkout

This module provides customers with a seamless shopping experience by enabling them to add items to their cart and proceed to checkout when they are ready.

Key Features:

- **Cart Management:** Customers can add or remove products from their cart, update quantities, and apply discount coupons.
 - **Checkout Process:** Streamlines the checkout process by collecting shipping details, selecting payment methods, and applying discounts or coupons.
 - **Cart Saving:** Saves the cart for future access if the customer does not complete the checkout process immediately.
 - **Order Summary:** Provides a detailed summary of the cart, including product details, total price, and shipping charges before the customer confirms the order.
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8. Discount and Offer Management

This module enables administrators and vendors to create, apply, and manage discounts and promotional offers for various products or customer segments.

Key Features:

- **Create Discounts:** Vendors and admins can create discount campaigns based on various criteria such as percentage discounts, flat rate discounts, or limited-time offers.
 - **Apply Coupons:** Customers can apply discount codes during checkout to receive reductions in price.
 - **Eligibility Rules:** Discounts can be applied based on conditions such as minimum order value, specific products, or user segments (e.g., first-time customers).
 - **Discount Tracking:** The system tracks which customers used discounts and how they affected sales.
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9. Feedback and Review Management

This module handles the collection and display of customer feedback and reviews on products and vendors.

Key Features:

- **Review Submissions:** Customers can submit reviews on products they've purchased, rating the product and providing comments.
- **Vendor Ratings:** Customers can rate their overall experience with a vendor.
- **Moderation:** Admins can moderate reviews to ensure they meet platform standards.
- **Display Reviews:** Displays reviews on product pages, helping other customers make informed decisions.

10. Analytics and Reporting Module

This module provides in-depth insights into the performance of the marketplace, including sales trends, customer behavior, and vendor performance.

Key Features:

- **Sales Reports:** Tracks overall sales, revenue generated, and the most popular products.
- **Customer Behavior Analytics:** Analyzes customer preferences, frequently purchased items, and abandoned carts.
- **Vendor Performance:** Provides insights into the top-performing vendors, product returns, and overall customer satisfaction.
- **Inventory Reports:** Helps vendors track their stock levels and identify products that need replenishment.
- **Custom Reports:** Admins and vendors can generate custom reports for specific time periods or product categories.

11. Notification Management

This module ensures that both customers and vendors receive relevant updates and notifications regarding their activities on the platform.

Key Features:

- **Order Notifications:** Customers receive real-time updates on their order status (e.g., shipped, delivered, etc.).
- **Vendor Alerts:** Vendors are notified about new orders, low stock levels, and customer reviews.
- **Promotional Notifications:** Sends out promotional offers, discounts, and newsletters to customers.
- **Return and Refund Alerts:** Notifies customers and vendors about return status updates and refunds processed.

These detailed modules will help guide the development of a comprehensive **Online Marketplace System** using Core Java, Collections, and Hibernate as the ORM tool. Each module is interconnected to ensure seamless functionality throughout the platform.

Detailed workflow

1. Vendor Management Workflow

Step-by-Step Flow:

1. Vendor Registration:

- Vendor visits the platform and chooses the "Register as Vendor" option.
- Vendor submits business details, including name, address, contact information, and business license.
- System verifies and stores the vendor's details in the database.
- Admin receives notification for approval of vendor registration.
- Once approved, the vendor receives a notification of successful registration.

2. Vendor Login:

- Vendor logs in with registered credentials.
- On successful authentication, the vendor is directed to their dashboard.

3. Product Management:

- Vendor adds products via the "Add Product" form.
- The product is categorized, and details such as price, stock, description, and images are uploaded.
- System stores product details and displays them to potential customers.

4. Order Tracking:

- Vendor receives notifications for new orders.
- Vendor checks order details and changes the order status (processing, shipped, delivered).
- System sends order status updates to the customer.

5. Sales and Analytics:

- Vendor can view sales performance, customer feedback, and reports on the dashboard.
- Reports can be generated based on time, products, and revenue.

2. Product Management Workflow

Step-by-Step Flow:

1. Product Addition:

- Vendor accesses the "Product Management" module.
- Vendor selects "Add Product" and provides necessary details (product name, category, description, price, quantity, and images).
- Product is stored in the database under the appropriate category.

2. Category Management (Admin):

- Admin can create, update, or delete categories.

- Admin logs in and navigates to the "Category Management" section.
- New categories can be added or existing categories can be modified.
- The system updates the product catalog to reflect category changes.

3. Stock Updates:

- Vendors monitor stock levels on their dashboard.
 - If a product is out of stock, the vendor updates the stock quantity.
 - Product availability is updated in real-time and reflected on the customer-facing storefront.
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3. Customer and Order Management Workflow

Step-by-Step Flow:

1. Customer Registration:

- Customer navigates to the "Register" page and submits personal details (name, email, address, password).
- System stores customer details in the database.
- Customer receives a confirmation email to verify their account.

2. Product Browsing:

- Customer browses the product catalog using search filters such as category, price range, and ratings.
- Customer adds items to the cart by selecting the "Add to Cart" button.

3. Order Placement:

- Customer reviews the cart and proceeds to checkout.
- At checkout, the customer selects the delivery address and applies any discount codes.
- Customer selects a payment method and confirms the order.
- The system generates an order and stores it in the database.
- Customer and vendor both receive notifications about the new order.

4. Order Processing:

- Vendor logs in to check new orders and begins processing them.
- Once the order is prepared, the vendor updates the order status to "Shipped."
- The customer receives shipment notifications and tracking details.

5. Order Delivery:

- Delivery service handles the shipment of the order.
- The system tracks the order's delivery progress.
- When the order is delivered, the system updates the status to "Delivered."

6. Order Return:

- If the customer is unsatisfied, they can initiate a return request.
- The vendor reviews the return request and either accepts or rejects it.

- If accepted, the system processes the refund and notifies the customer.
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4. Payment Gateway Workflow

Step-by-Step Flow:

1. Payment Initiation:

- After reviewing the cart, the customer selects a payment method (credit card, debit card, or e-wallet).
- The customer is redirected to a secure payment gateway page.

2. Payment Processing:

- Customer enters payment details and submits the transaction.
- The payment gateway verifies the transaction and confirms payment with the system.

3. Invoice Generation:

- On successful payment, the system generates an invoice.
- The invoice is stored in the database and emailed to the customer for their records.

4. Refund Processing (If Applicable):

- If the customer cancels or returns an order, the system initiates a refund request through the payment gateway.
 - Once the refund is processed, both customer and vendor receive a notification.
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5. Shipment Tracking Workflow

Step-by-Step Flow:

1. Order Shipment:

- Vendor packs the product and arranges shipping via an integrated logistics service.
- Vendor updates the order status to "Shipped" and provides the shipment tracking number.

2. Tracking Updates:

- The customer receives a shipment notification with tracking details.
- The system periodically updates the tracking status (in transit, out for delivery, delivered).

3. Delivery Confirmation:

- Once the order is delivered, the system updates the order status to "Delivered."
 - Customer receives a delivery confirmation message.
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6. Discount and Offer Management Workflow

Step-by-Step Flow:

1. Discount Creation:

- Admin or vendor creates a new discount by specifying the discount type (percentage or flat rate) and the eligibility criteria (specific products, minimum order value, etc.).

- The system applies the discount to the relevant products or orders.

2. Coupon Code Application:

- During checkout, customers can apply discount coupons to their orders.
- The system validates the coupon code and applies the discount if eligible.

3. Sales Tracking:

- The system tracks which discounts were used by customers and generates reports on discount effectiveness for the admin and vendors.

7. Feedback and Review Management Workflow

Step-by-Step Flow:

1. Customer Review Submission:

- After receiving an order, the customer can navigate to the order history and select the "Leave Review" option.
- Customer rates the product and writes a review.
- The review is stored in the system and associated with the product and vendor.

2. Review Moderation:

- Admin reviews the submitted feedback to ensure compliance with platform guidelines.
- Approved reviews are displayed on the product page for other customers to view.

3. Vendor Feedback:

- Vendors can view customer feedback and ratings through their dashboard.
- The feedback helps vendors improve their services or resolve customer complaints.

8. Analytics and Reporting Workflow

Step-by-Step Flow:

1. Sales Reports:

- Vendors can generate sales reports based on specific time periods or product categories.
- The system aggregates sales data and generates visual reports (graphs, charts).

2. Customer Behavior Tracking:

- The system tracks customer interactions (e.g., product views, search queries, abandoned carts).
- Admins can analyze customer behavior trends to make informed decisions on product listings and discounts.

3. Vendor Performance Analytics:

- Vendors can view reports on top-performing products, sales trends, and customer satisfaction scores.

9. Notification Management Workflow

Step-by-Step Flow:

1. Order Notifications:

- The system sends real-time notifications to customers about order placement, shipment status, and delivery.

2. Promotional Notifications:

- The system sends notifications about special offers, discounts, and new product arrivals to registered customers.

3. Vendor Alerts:

- Vendors receive notifications for new orders, product reviews, low stock levels, and payments.

These workflows help define the processes for the **Online Marketplace System**, providing a clear understanding of how users and entities will interact across the different modules.